Learned Complementary Preconditioning for RBF-FD Discretizations

Qimu Luo

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Abstract

This note investigates learned complementary preconditioning for linear systems arising from radial basis function-generated finite difference (RBF-FD) discretizations of partial differential equations. The starting point is a strong classical RBF-FD preconditioner. The main objective is to characterize the inverse action that remains unresolved by this classical method and determine whether it can be exploited by machine learning. Two learned constructions are considered. The first directly learns the missing correction, while the second first identifies a low-dimensional difficult subspace and learns how this subspace changes across related RBF-FD systems. In the latter case, the predicted subspace is used in a structured coarse-space correction. The two learned approaches will be compared rather than assuming in advance that one is superior.

1 Introduction and Problem Formulation

Consider a standard elliptic boundary-value problem

$$-\Delta u = f \qquad \text{in } \Omega, \tag{1}$$

$$u = g \qquad \text{on } \partial\Omega, \tag{2}$$

where $\Omega \subset \mathbb{R}^d$. The initial study will use this problem as a controlled setting before considering more general differential operators or variable coefficients.

Let

$$X = \{x_1, \dots, x_N\}$$

be a set of nodes in $\overline{\Omega}$. For each interior node x_i, RBF-FD selects a local stencil

$$S_i = \{x_{i_1}, \dots, x_{i_s}\}.$$

The differential operator is approximated locally by

$$-\Delta u(x_i) \approx \sum_{j=1}^{s} w_{ij} u(x_{i_j}), \tag{3}$$

where the weights w_{ij} are constructed from radial basis functions, usually together with polynomial augmentation. Boundary rows are imposed directly from the Dirichlet condition. Collecting all stencil equations gives

$$A_{\mathrm{RBF}} u = b, \tag{4}$$

where $u \in \mathbb{R}^N$ contains the nodal values of the numerical solution.

Unlike global RBF collocation, RBF-FD uses local stencils and normally produces a sparse matrix. This makes sparse preconditioning and Krylov iterative methods natural tools for solving (4). The numerical properties of A_{RBF} depend on the stencil size, kernel, polynomial degree, node geometry, resolution, and differential operator. The first stage of the study will therefore use RBF-FD throughout rather than mixing global and local RBF formulations. Global RBF collocation may be considered later as a separate extension [8].

2 Classical Preconditioning and the Missing Inverse Action

Let u^* denote the exact solution of the discrete system

$$A_{\mathrm{RBF}} u^* = b.$$

At iteration m, define

$$e^{(m)} = u^* - u^{(m)}, \tag{5}$$

$$r^{(m)} = b - A_{\mathrm{RBF}} u^{(m)}. \tag{6}$$

Since $A_{\mathrm{RBF}} u^* = b$,

$$r^{(m)} = A_{\mathrm{RBF}} e^{(m)}, \tag{7}$$

and therefore

$$e^{(m)} = A_{\mathrm{RBF}}^{-1} r^{(m)}. \tag{8}$$

Thus, $A_{\mathrm{RBF}}^{-1} r^{(m)}$ is the exact correction associated with the current residual.

Let M_{RBF} denote a classical RBF-FD preconditioner. In a stationary interpretation,

$$u^{(m+1)} = u^{(m)} + M_{\mathrm{RBF}}^{-1} r^{(m)}. \tag{9}$$

The corresponding error propagation is

$$e^{(m+1)} = \left(I - M_{\mathrm{RBF}}^{-1} A_{\mathrm{RBF}}\right) e^{(m)}. \tag{10}$$

Define

$$S_{\mathrm{RBF}} = I - M_{\mathrm{RBF}}^{-1} A_{\mathrm{RBF}}. \tag{11}$$

If S_{RBF} is diagonalizable and

$$S_{\mathrm{RBF}} v_i = \mu_i v_i,$$

then

$$e^{(m)} = \sum_i \alpha_i^{(m)} v_i \tag{12}$$

implies

$$e^{(m+1)} = \sum_i \mu_i \alpha_i^{(m)} v_i. \tag{13}$$

Components associated with small $|\mu_i|$ are strongly damped, while components associated with larger $|\mu_i|$ may persist. This motivates the idea of a difficult subspace.

For a nonsymmetric or nonnormal RBF-FD matrix, eigenvectors alone may not characterize convergence. Singular vectors, Ritz information, residual histories, and data-driven subspaces will therefore also be considered.

The main quantity of interest is

$$\Delta_{\mathrm{RBF}}(r) = A_{\mathrm{RBF}}^{-1} r - M_{\mathrm{RBF}}^{-1} r. \tag{14}$$

It measures the difference between the exact correction and the correction supplied by the classical preconditioner. The central numerical question is whether this unresolved correction has sufficiently regular structure to justify a learned complementary method.

3 Classical Baseline

A strong existing RBF-FD preconditioner will first be selected and implemented. Since RBF-FD matrices are sparse, natural candidates include incomplete-factorization methods and RBF-FD-specific smaller-stencil preconditioners [8]. The classical method is not fixed in advance solely because it is convenient to implement; it will be chosen according to the observed structure of A_{RBF} and existing RBF-FD preconditioning results. The classical baseline has two purposes:

(i) establish the convergence and computational cost without machine learning;

(ii) generate the unresolved correction (14) that the learned methods will attempt to exploit.

4 Structure of the Unresolved Correction

For a fixed RBF-FD system, consider residuals

$$r^{(1)}, \ldots, r^{(K)}$$

and define

$$d^{(j)} = \Delta_{\mathrm{RBF}}(r^{(j)}) = A_{\mathrm{RBF}}^{-1} r^{(j)} - M_{\mathrm{RBF}}^{-1} r^{(j)}. \tag{15}$$

Collect the unresolved corrections into

$$D = \begin{bmatrix} d^{(1)} & d^{(2)} & \cdots & d^{(K)} \end{bmatrix} \in \mathbb{R}^{N \times K}. \tag{16}$$

Its singular value decomposition is

$$D = U \Sigma V^T, \qquad \sigma_1 \geq \sigma_2 \geq \cdots \geq 0. \tag{17}$$

For a chosen dimension q, let

$$Z_q = \begin{bmatrix} u_1 & \cdots & u_q \end{bmatrix}. \tag{18}$$

The space range(Z_q) is the POD subspace associated with the unresolved corrections. By the Eckart–Young–Mirsky theorem,

$$\left\| D - Z_q Z_q^T D \right\|_F^2 = \sum_{i>q} \sigma_i^2. \tag{19}$$

The fraction of captured snapshot energy is

$$E_q = \frac{\sum_{i=1}^{q} \sigma_i^2}{\sum_i \sigma_i^2}. \tag{20}$$

If E_q is close to one for $q \ll N$, this provides evidence that the sampled unresolved corrections are approximately low-dimensional.

This conclusion must also hold outside the training snapshots. For a new unresolved correction d_{test}, define

$$\eta_q = \frac{\| (I - Z_q Z_q^T) d_{\mathrm{test}} \|_2}{\| d_{\mathrm{test}} \|_2}. \tag{21}$$

A small value of η_q indicates that the POD space also represents unseen unresolved corrections.

POD is only a diagnostic and reference method. It does not establish that the same subspace is optimal for Krylov convergence. The identified space will therefore also be evaluated by its effect on iteration counts and residual histories. Ritz vectors, Krylov information, or other subspace constructions may be considered if they better represent the components that actually slow the iterative solver.

5 Two Learned Complementary Approaches

The study will compare two learned strategies. Both retain the classical preconditioner M_{RBF}^{-1} and use machine learning only for the part that it does not resolve well.

5.1 Direct Learned Complement

The first approach learns the unresolved correction directly:

$$\mathcal{N}_\theta(r) \approx \Delta_{\mathrm{RBF}}(r) = A_{\mathrm{RBF}}^{-1} r - M_{\mathrm{RBF}}^{-1} r. \tag{22}$$

A basic supervised objective is

$$\mathcal{L}_{\mathrm{dir}}(\theta) = \mathbb{E}\left[\left\| \mathcal{N}_\theta(r) - \left(A_{\mathrm{RBF}}^{-1} r - M_{\mathrm{RBF}}^{-1} r \right) \right\|_2^2 \right]. \tag{23}$$

The resulting hybrid correction is

$$\mathcal{P}_{\mathrm{dir}}(r) = M_{\mathrm{RBF}}^{-1} r + \mathcal{N}_\theta(r). \tag{24}$$

This approach makes the weakest structural assumption. It does not require the unresolved correction to lie in a low-dimensional subspace. Its main disadvantages are that the network must predict a full N-dimensional correction and the resulting preconditioning action may be nonlinear or iteration-dependent.

5.2 Learned Difficult-Subspace Complement

The second approach is used only if the analysis in Section 4 provides evidence that the unresolved correction is approximately low-dimensional. Suppose

$$\mathcal{V}_{\mathrm{diff}} \approx \mathrm{range}(Z), \qquad Z \in \mathbb{R}^{N \times q}, \qquad q \ll N.$$

A correction restricted to this space is

$$\delta u_c = Z\alpha.$$

For the symmetric case, imposing the Galerkin condition gives

$$Z^T \left(r - A_{\mathrm{RBF}} Z\alpha \right) = 0, \tag{25}$$

so that

$$Z^T A_{\mathrm{RBF}} Z\alpha = Z^T r. \tag{26}$$

Provided $Z^T A_{\mathrm{RBF}} Z$ is nonsingular,

$$\delta u_c = Z \left(Z^T A_{\mathrm{RBF}} Z \right)^{-1} Z^T r. \tag{27}$$

For nonsymmetric systems, the corresponding Petrov–Galerkin or deflation formulation can be used without changing the main research idea.

To make the coarse correction complementary to the classical preconditioner, first compute

$$y_0 = M_{\mathrm{RBF}}^{-1} r, \tag{28}$$

and then form the remaining residual

$$r_1 = r - A_{\mathrm{RBF}} y_0 = \left(I - A_{\mathrm{RBF}} M_{\mathrm{RBF}}^{-1} \right) r. \tag{29}$$

The two-level correction is

$$\mathcal{P}_Z(r) = M_{\mathrm{RBF}}^{-1} r + Z \left(Z^T A_{\mathrm{RBF}} Z \right)^{-1} Z^T \left(I - A_{\mathrm{RBF}} M_{\mathrm{RBF}}^{-1} \right) r. \tag{30}$$

For a fixed system, Z can be obtained from POD or another diagnostic method. No neural network is needed if this fixed space remains effective. Machine learning is introduced only if the useful difficult subspace changes across related RBF-FD systems.

6 Learning the Difficult Subspace Across RBF-FD Systems

Let ξ describe an RBF-FD system. It may contain information such as

$$\xi = (\varepsilon, X, N, s, \text{kernel}, \text{PDE parameters}),$$

where s denotes the stencil size. Write

$$A(\xi)u = b, \qquad M(\xi)^{-1}$$

for the corresponding linear system and classical preconditioner.

For each training system ξ_j, let $Z_*(\xi_j)$ denote a reference difficult-subspace basis obtained from POD or another subspace-identification procedure. The learned task is

$$\xi \longmapsto \text{range}\left(Z_*(\xi)\right). \tag{31}$$

A subspace basis is not unique. If $Q \in \mathbb{R}^{q\times q}$ is orthogonal, then Z and ZQ span the same space. A direct basis-wise loss is therefore not appropriate. For orthonormal bases, define

$$\Pi_\theta = Z_\theta Z_\theta^T, \qquad \Pi_* = Z_* Z_*^T. \tag{32}$$

A basis-invariant loss is

$$\mathcal{L}_{\text{sub}}(\theta) = \mathbb{E}_\xi\left[\|\Pi_\theta(\xi) - \Pi_*(\xi)\|_F^2\right]. \tag{33}$$

Principal-angle-based losses may also be considered.

Because the number and locations of RBF-FD nodes may vary between systems, the learned model should not require a fixed-dimensional vector representation. Graph-based architectures are natural candidates because RBF-FD already provides local stencil connectivity. Operator-learning architectures are another possible choice. The model architecture will be selected after the transferability of the difficult subspaces has been established.

Given a predicted basis Z_θ, the correction is still computed through the reduced numerical solve:

$$\mathcal{C}_\theta(r) = Z_\theta\left(Z_\theta^T A(\xi) Z_\theta\right)^{-1} Z_\theta^T r. \tag{34}$$

The resulting hybrid preconditioning action is

$$\mathcal{P}_{\text{sub}}(r) = M(\xi)^{-1} r + \mathcal{C}_\theta\left(I - A(\xi)M(\xi)^{-1}\right) r. \tag{35}$$

7 Methodology and Research Plan

The study will proceed in the following stages.

Stage 1: RBF-FD discretization and classical baseline.

A standard elliptic PDE with Dirichlet boundary conditions will first be discretized by RBF-FD. The matrix structure will be examined over different node sets, resolutions, stencil sizes, and RBF parameters. A strong classical RBF-FD preconditioner will then be selected and implemented.

Stage 2: Unresolved-correction analysis.

Reference solutions will be used to generate

$$d = A(\xi)^{-1} r - M(\xi)^{-1} r. \tag{36}$$

Singular-value decay, captured energy, out-of-sample projection error, and Krylov convergence will be used to test whether the unresolved correction is approximately low-dimensional.

Stage 3: Transferability analysis.

If low-dimensional structure is observed, the corresponding subspaces will be compared across related RBF-FD systems. Projector differences and principal angles will be used to determine whether the difficult subspace changes systematically with ξ.

Stage 4: Direct learned complement.

A learned model will be trained using (23). This provides a general learned-preconditioning baseline that does not assume low-dimensional structure.

Stage 5: Learned subspace complement.

If the low-dimensional and transferability hypotheses are supported, a graph-based or operator-learning model will be trained using (33). The predicted subspace will then be used in the structured correction (35).

Stage 6: Krylov solver integration.

The resulting preconditioners will be embedded in a Krylov solver. GMRES is appropriate when the preconditioning action is fixed and linear for a nonsymmetric system. If the learned preconditioner changes during the solve or depends nonlinearly on the current residual, FGMRES is more appropriate [3].

8 Evaluation

The main numerical comparison will include:

(i) classical RBF-FD preconditioning;

(ii) classical preconditioning with generic neural post-correction;

(iii) classical preconditioning with a POD coarse-space correction;

(iv) classical preconditioning with a direct learned complement;

(v) classical preconditioning with a learned difficult-subspace correction.

The direct learned complement and learned subspace complement answer different questions. The former tests whether the missing inverse action can be learned directly. The latter tests whether exploiting an explicit low-dimensional structure improves accuracy, transferability, stability, or computational cost. The main solver metrics are:

- Krylov iteration count;
- residual reduction and convergence rate;
- robustness across RBF-FD parameters and node sets;
- preconditioner setup and inference cost;
- total wall-clock time.

For the subspace-based methods, additional diagnostics include:

- singular-value decay of the unresolved-correction snapshots;
- captured energy E_q;

- out-of-sample projection error η_q;
- principal angles between difficult subspaces;
- sensitivity to the coarse dimension q.

Generalization will be tested across variations in ε, N, node geometry, stencil size, kernel choice, and PDE parameters.

Because machine learning introduces offline and inference costs, the total computational cost for repeated related solves will be measured by

$$T_{\text{total}} = T_{\text{train}} + T_{\text{setup}} + N_{\text{solve}} T_{\text{online}}. \tag{37}$$

A learned preconditioner is useful only if the reduction in online solution cost compensates for its additional training, prediction, and setup costs.

9 Research Position and Expected Contribution

Learned preconditioning and neural enhancement of iterative solvers are established research areas. Existing approaches include learned factorizations, DeepONet-based preconditioning, Krylov-aware neural preconditioners, structure-preserving neural methods, and learned coarse spaces [11, 9, 5, 3, 6, 12]. Related work combines neural operators with relaxation and multigrid methods [13, 7, 2, 10]. Classical preconditioning for RBF and RBF-FD systems is also well established [4, 1, 8].

The proposed study therefore does not treat either neural preconditioning or coarse-space correction as a new idea by itself. Its focus is the unresolved action of an RBF-FD-specific classical preconditioner. The first objective is to determine whether this unresolved correction has reproducible structure. The second is to compare two ways of exploiting that structure: direct learning of the missing inverse action and learning of a transferable difficult subspace. The third is to determine whether either learned construction provides a measurable advantage over strong classical and POD-based baselines.

The central research questions are:

(i) Does the inverse action left unresolved by a classical RBF-FD preconditioner exhibit low-dimensional or otherwise learnable structure?

(ii) Is this structure transferable across related RBF-FD discretizations?

(iii) Is direct learning of the missing inverse action or learned subspace correction more effective in terms of convergence, robustness, and total computational cost?

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